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Project 3 Report

Introduction

For this project, our overall question was “What distinguishes successful Steam games from the rest of the Steam catalog?” Steam is one of the largest digital marketplaces for games, providing a useful environment for examining the factors associated with game popularity and player engagement. In this project, we investigated what distinguishes highly successful games from the broader Steam ecosystem and whether characteristics such as price, reviews, ratings, and player engagement are associated with success. We created a dataset of approximately 20,000 Steam games by querying the Steam Web API. The resulting dataset contained over 150 variables describing characteristics such as player counts, reviews, ratings, prices, and game attributes.

We used exploratory data analysis, statistical comparisons, correlation analysis, and multiple linear regression to investigate patterns in the data. Our results showed that higher priced games tend to be more successful, providing insight into how players view their games and libraries as investments. We also determined that player engagement is highly concentrated. Approximately 69% of sampled games had no average 30-day players, while only 3.1% had more than 100,000. Games in the highest-success group generally had substantially more reviews and recommendations, as well as higher review and Metacritic scores. Regression analysis found that basic game characteristics combined with review counts and recommendations increased the explanatory power of the model to approximately 67%. These findings provide insight into the characteristics associated with successful games and demonstrate the strong relationship between a game's existing community engagement and its current player base

To view our Project 3 presentation, follow this link: [📺 CS5830-Project3](#) . And to see our data analysis, follow this link to our github: [ewalburger/CS5830_Project3](https://github.com/ewalburger/CS5830_Project3). To see your dataset, follow this link to box: <https://usu.box.com/s/fm4pbnddcq7uzugsrqcp8g6s8fcuemo7>.

Dataset

For this project, we created our own dataset by querying the Steam Web API. We had a limited amount of queries, which naturally determined the size of the dataset. Each query returned 500 games, and we queried the API 40 times resulting in a total of 20,000 games and their data for our set. 20,000 games was a sufficient sample of games on Steam for the types of analyses we wanted to do. We formatted the raw json data into a csv file by reading the json into a dataframe, and then writing it to a csv that we could easily parse and query. Our dataset had 179 columns for each game row. These columns included information about ratings, game costs in different countries, player trends, and attributes of the games.

The dataset is suitable for analysis because the data is accurate and being updated in real-time from Steam. We gathered a large enough sample to be sufficiently representative of overall game trends. The values in each column are consistent data types and values, and it is easy to remove NaN values without losing lots of information. There are also no duplicate games, so each game in our dataset is unique.

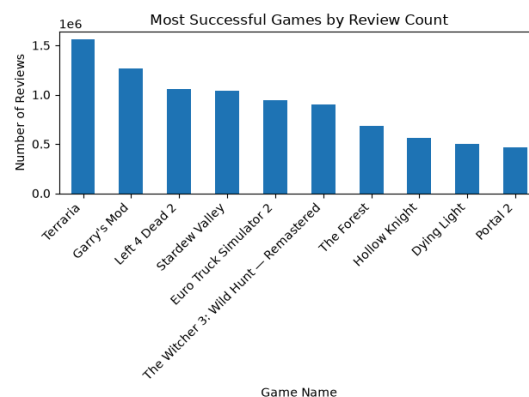
Analysis Technique

We used several different techniques to analyze our dataset. We filtered and sorted data using “nlargest” to find distinctions between the most successful and least successful games. We also used quantiles to find specific percentiles to find cutoffs for the most successful game category. In our analysis of price distributions we compared averages and ran the independent t-test to see if there’s a statistically significant difference between the average costs of successful vs less successful games. We used boxplots to show cost distributions, and bar charts to represent the differences in numbers of reviews across successful games.

We also examined the distribution of player engagement using histograms and summary statistics with some transformations to account for skewed player-count data. We compared highly successful games with other games using median values and boxplots, and used correlation analysis to examine relationships between player engagement and other game characteristics. We used multiple linear regression to evaluate which characteristics were most strongly associated with player engagement while considering several variables simultaneously.

Results

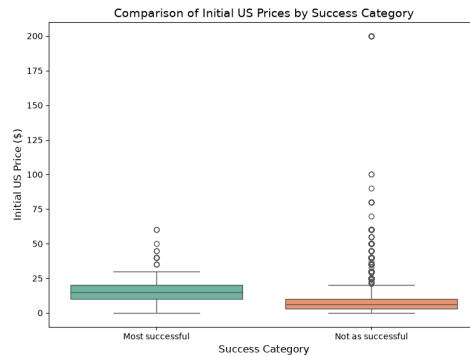
The success of a game can be measured in many different ways. One way we found to distinguish the most successful games from the rest was to look at the top rated games with the most reviews. Because the game data included the number of total reviews and the overall rating, which summarizes the percentage of positive reviews as an index (0-9), we were able to get a comprehensive idea of what games lots of people really enjoy. We generated a bar-chart of the ten most highly rated games with the highest number of reviews. This data is valuable because it could help inform Steam on which games the most people really enjoy, which is important for giving players game recommendations.



Because we wanted to look at some of the characteristics that set the most successful games apart from the rest, we needed to define a cutoff when looking at the highest rated games. We chose to group the top 90th percentile of total reviews for games with a rating of 9, which was the highest rating, as the most successful games. This left us with 272 most successful games and 19,728 not so successful games.

We then analyzed the average price differences and cost distributions depending on game success. We found that the most successful games had an average initial price of \$15.44, while the not as successful games had an average initial price of \$8.75. There is a statistically significant difference in the average initial prices between the most successful games and the not as successful games with a T-statistic of 10.2990 and a p-value of 3.0008e-21. We represented the distribution of these differences using a box and whisker plot (see below). This

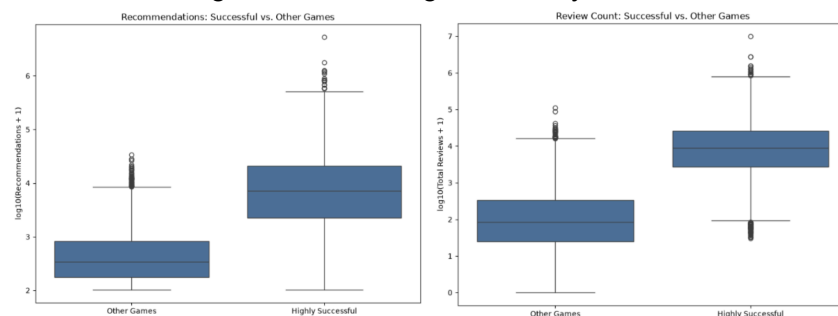
data could be very interesting to stakeholders in different games. We thought that the most popular games would cost less on average, which would make them more accessible to more people. We also thought lower prices would be more attractive to people, which would make them rate the games more highly. However, it does make sense that people would rate high quality games more highly, and high quality games can be sold for more money.



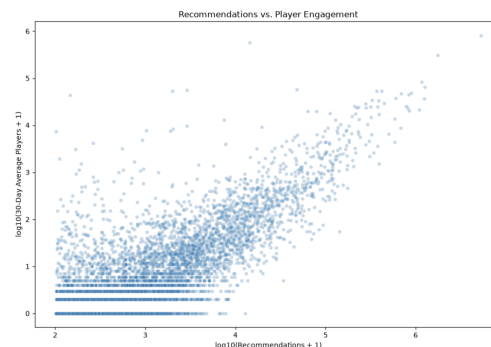
We decided to also check what kinds of games kept players engaged. We noticed that player engagement was highly concentrated across the Steam ecosystem. From the games sampled, 69% of games had 0 average 30-day players while only 3.1% had more than 100,000. We defined “highly successful” games as the top 10% of games which had a 30-day average player count of 16 or more. This should give warning to stakeholders attempting to develop successful games that not many people are able to do so.

	name	avg30d	peak30d	peakalltime
	Counter-Strike 2	811711.0	1452036.0	1818368.0
	Dota 2	574908.0	919002.0	1291328.0
	PUBG: BATTLEGROUNDS	310923.0	788224.0	3236027.0
	Rust	83242.0	159010.0	259646.0
	Tom Clancy's Rainbow Six Siege	64298.0	120267.0	200476.0
	Team Fortress 2	57205.0	90404.0	253225.0
	Warframe	55840.0	108836.0	181509.0
	Project Zomboid	53429.0	100244.0	124506.0
	The Binding of Isaac: Rebirth	53111.0	92691.0	155966.0
	War Thunder	52907.0	100117.0	125317.0

Looking at these games, we note that they differed substantially from others across several characteristics. Median review count was nearly 9000 compared with 84 from the other group, and the median recommendations were 7255 compared with 337. Successful games have higher median Metacritic scores and review scores, with statistical tests indicating significant differences across these characteristics. Individuals or companies wanting to promote a game should know that recommending it and reviewing it are likely factors to increasing its popularity.



A regression analysis found that basic game characteristics alone explained 16% of the variation in player engagement, while adding review counts and recommendations increased the model's explanatory power to 67%. Review count and recommendations were the strongest predictors in the expanded model. The correlation analysis similarly showed a positive relationship between player engagement and Metacritic scores, recommendations, and review counts. This indicates that a game's existing popularity and community engagement are strongly associated with its current player base. Developers seeking to create popular games should know that not only is a popular game with an engaged player base created, but also maintained. Those seeking fun experiences with large online groups should know that while new games are cool, older games are reliable for these experiences.



Technical

The raw Steam Web API data was converted from JSON into a CSV for analysis. We cleaned and prepared the data by examining missing values, checking variable types, removing or excluding missing values where necessary, and creating derived variables such as `price_usd` to account for the currency exchanges provided for us and `log_avg30d` to account for the extreme right-skew in player counts. The API limited the number of queries available to us, so our dataset consisted of about 20,000 games rather than the entire Steam catalog. We had trouble interpreting some of the column definitions because the documentation the API provided was scant and so we had to make some inferences. We initially encountered unexpected concentrations of games with zero `avg30d` values, but we found that many of these games had historical player activity, so we retained these observations rather than removing the data. We used quantiles to define highly successful games based on player engagement, descriptive statistics and boxplots to compare successful and other games, Welch's independent t-test to compare price distributions, and correlation and multiple linear regression to examine relationships between game characteristics and player engagement. Regression was appropriate because our goal was to evaluate the simultaneous association of multiple game characteristics with a continuous engagement measure. We also compared models with and without review counts and recommendations to determine how much explanatory power these popularity-related variables added. Alternative approaches could have included using a larger or randomly sampled Steam dataset, different definitions of success, or classification techniques, but our available API queries and very recent introduction to some of the aforementioned techniques led us to focus on descriptive and regression-based methods.